

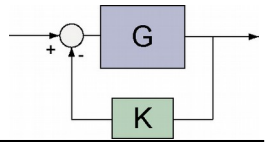
ME 2801

Introduction to Control Systems

W8: Frequency Response,
Bode Plot Introduction
Part 2



Today



Bode Plot Building Blocks

- Gain
- Derivatives/Integrals
- First-Order
- Second Order

Sketching Process and Example

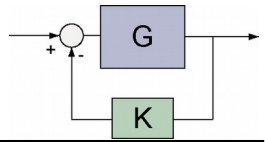
- Graphical summation of individual building blocks

Announcements

- Bodepaper and bodesketch programs posted on course website – see [Resources Page](#).
- Swarthmore has a site that might be useful...
<http://lpsa.swarthmore.edu/Bode/Bode.html>



Building Block #1: Gain



Trans. Func.

$$G(s) = K$$

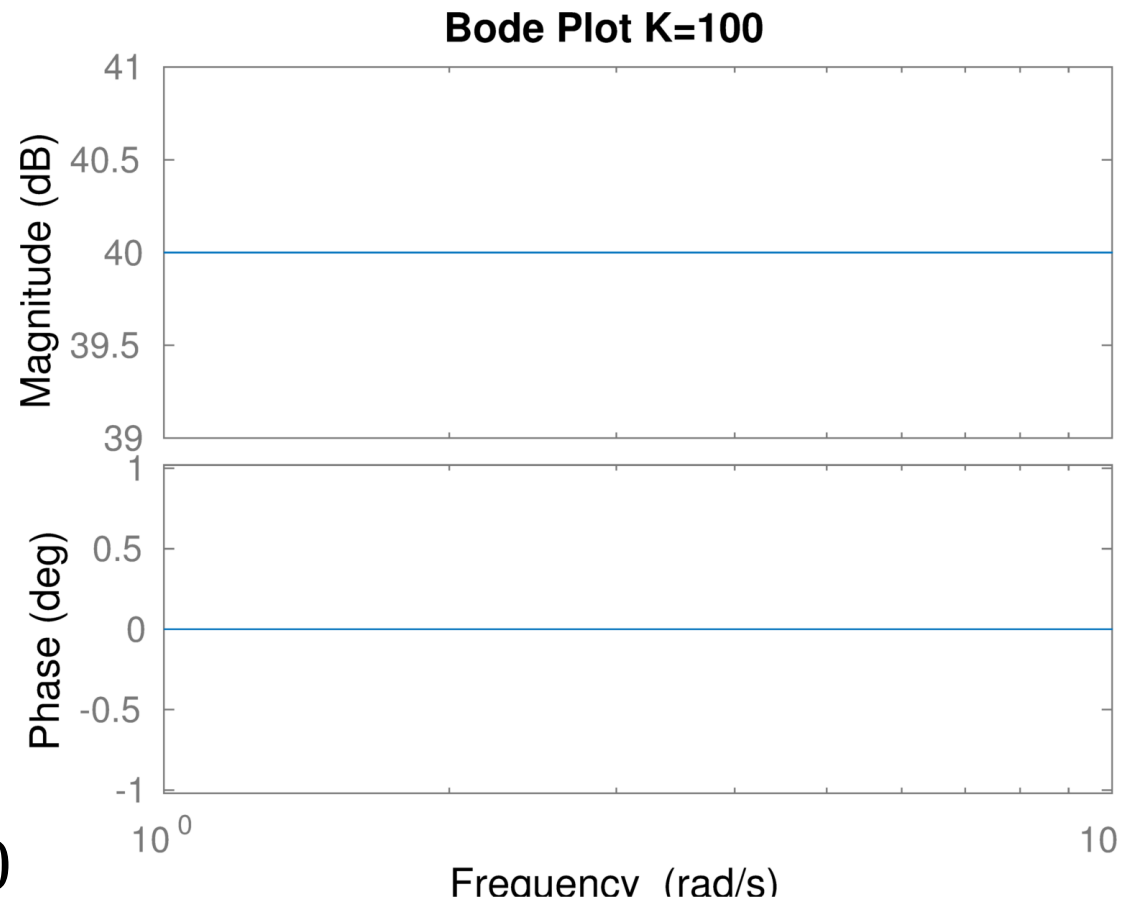
Magnitude

$$|G(s = j\omega)| = K$$

$$M = 20 \log_{10}(K) = \text{constant}$$

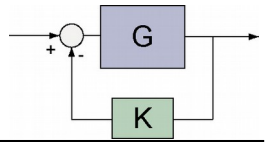
Phase

$$\angle G(s = j\omega) = \tan^{-1} \left[\frac{\Im(G)}{\Re(G)} \right] = 0$$





Building Block #1: Gain



Trans. Func.

$$G(s) = K$$

Magnitude

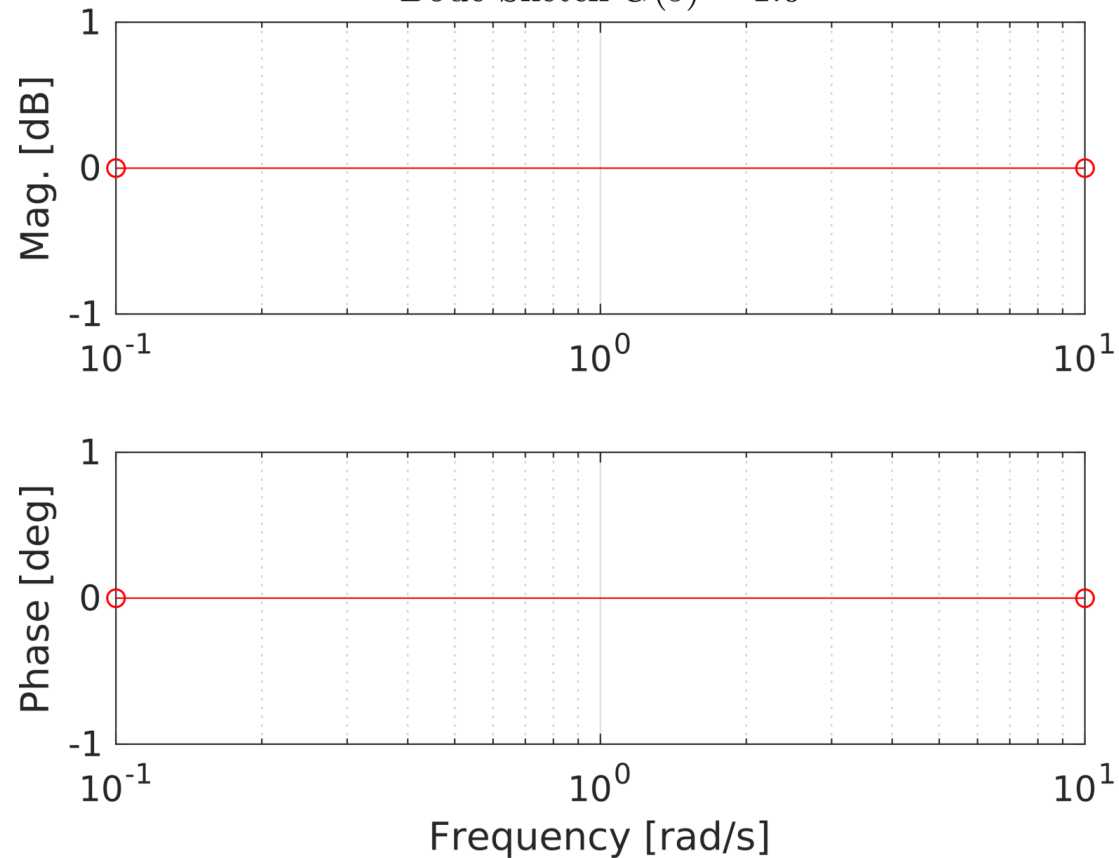
$$|G(s = j\omega)| = K$$

$$M = 20 \log_{10}(K) = \text{constant}$$

Phase

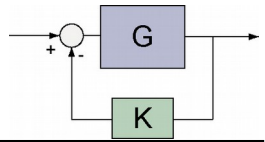
$$\angle G(s = j\omega) = \tan^{-1} \left[\frac{\Im(G)}{\Re(G)} \right] = 0$$

Bode Sketch $G(s) = 1.0$





Building Block #1: Gain



Trans. Func.

$$G(s) = K$$

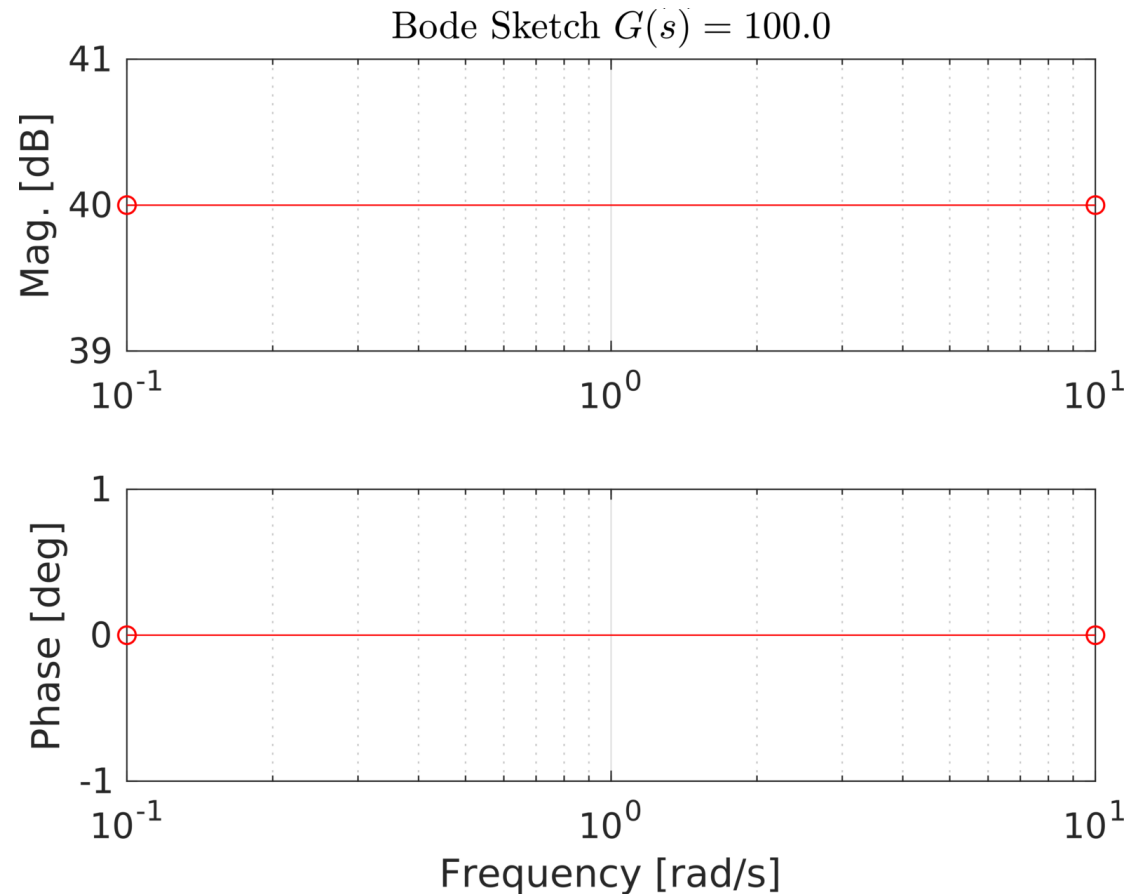
Magnitude

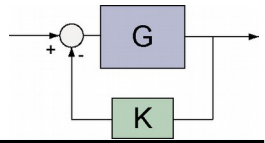
$$|G(s = j\omega)| = K$$

$$M = 20 \log_{10}(K) = \text{constant}$$

Phase

$$\angle G(s = j\omega) = \tan^{-1} \left[\frac{\Im(G)}{\Re(G)} \right] = 0$$





Derivatives and Integrals

Derivative, first-order

$$G(s) = s$$

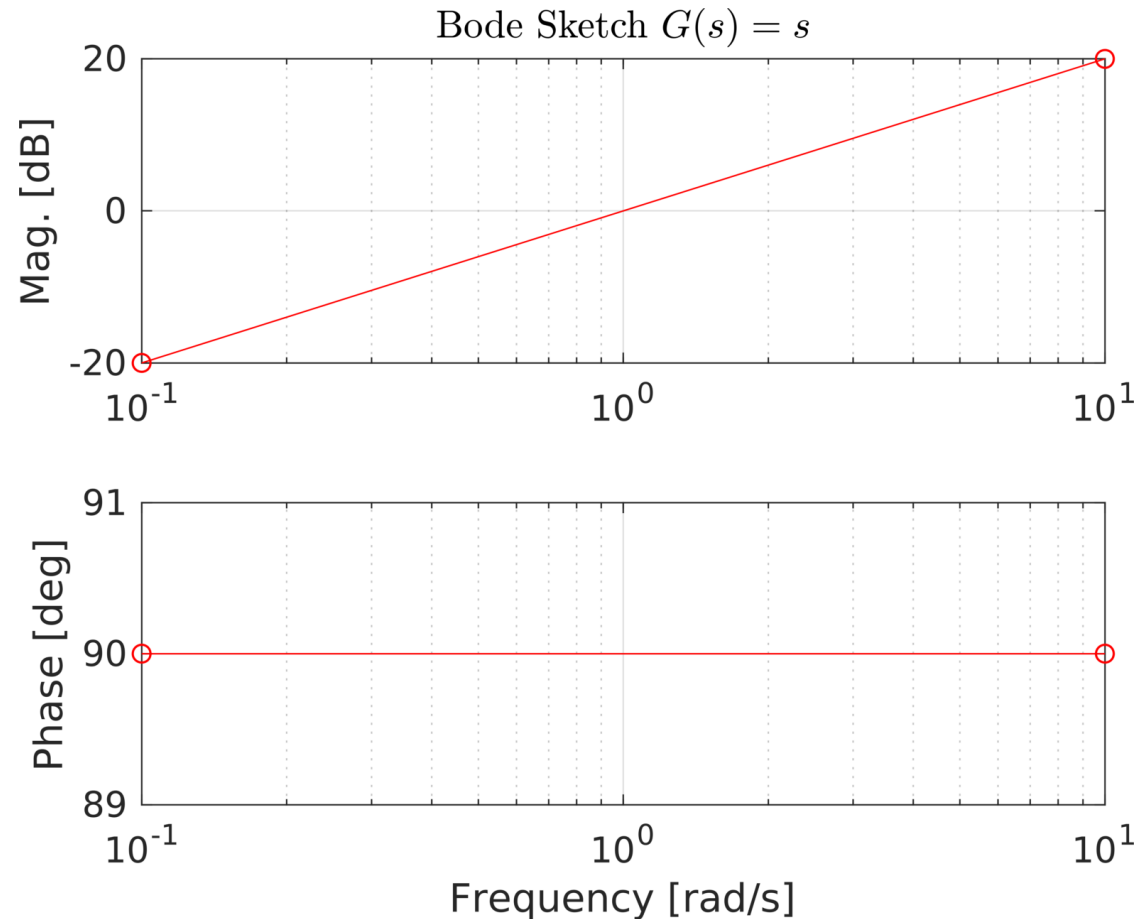
Magnitude

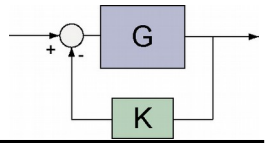
$$M = |G(s = j\omega)| = \omega$$

- Slope of “20 dB/decade”

Phase

$$\angle G(s = j\omega) = +90$$





Derivatives and Integrals

Derivative, first-order

$$G(s) = s, 10s, 0.1s$$

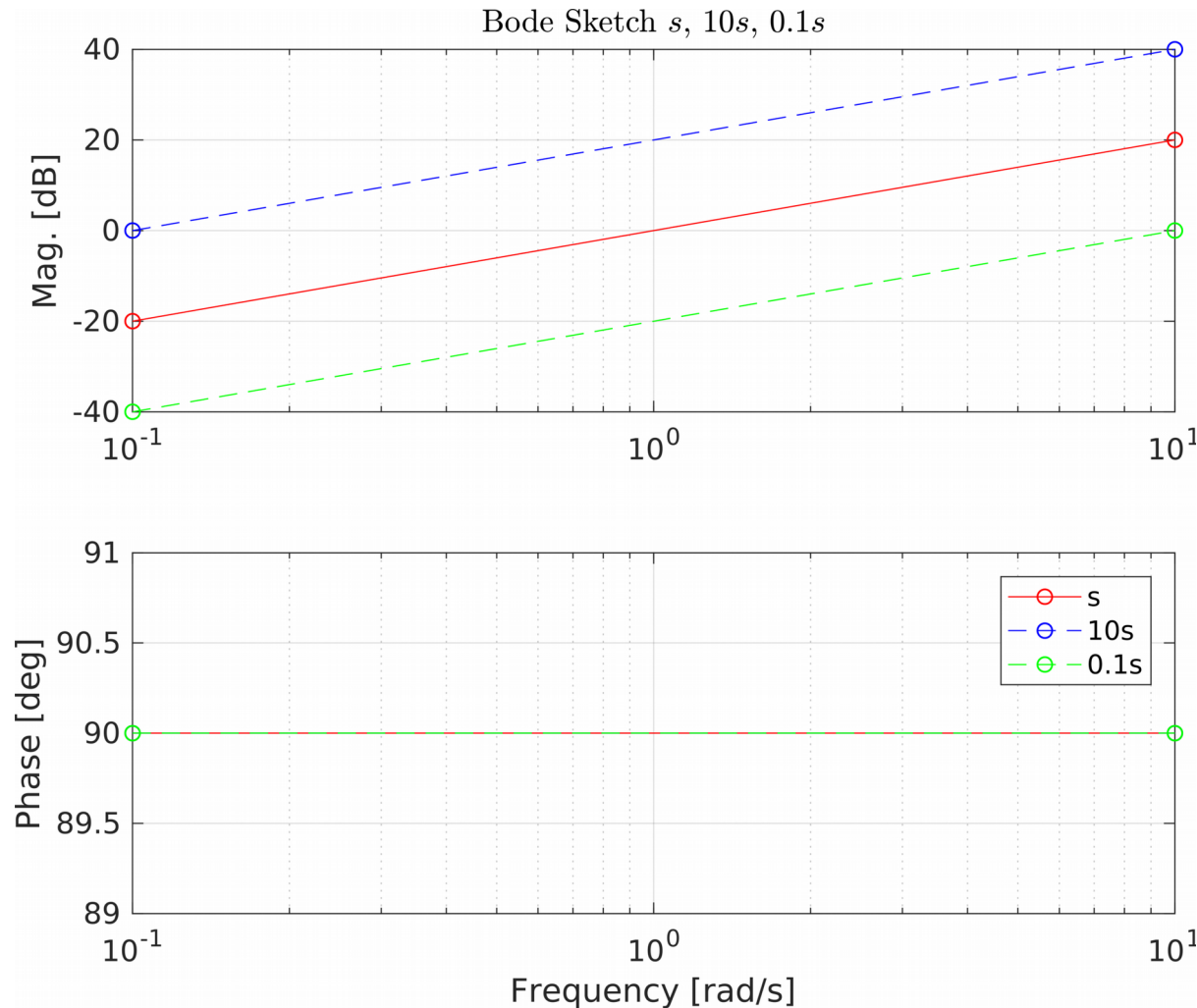
Magnitude

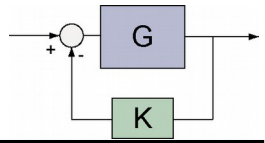
$$M = |G(s = j\omega)| = \omega$$

- Slope of “20 dB/decade”

Phase

$$\angle G(s = j\omega) = +90$$





Derivatives and Integrals

Derivative, first-order

$$G(s) = s^n$$

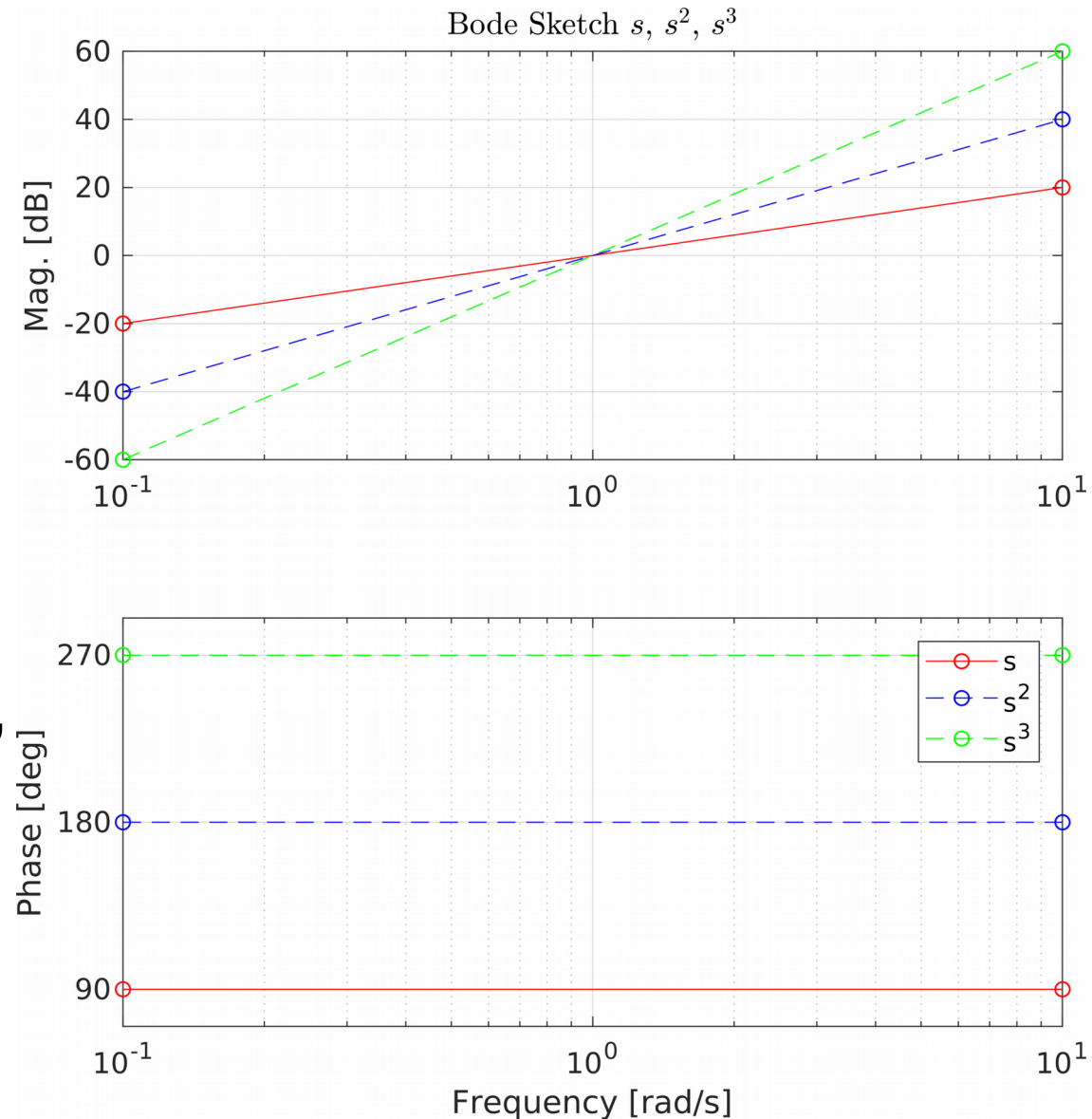
Magnitude

$$M = |G(s = j\omega)| = \omega^n$$

- Slope of “20*n dB/decade”

Phase

$$\angle G(s = j\omega) = +90 * n$$



Derivatives and Integrals

Order -1 (integral)

$$G(s) = \frac{1}{s}$$

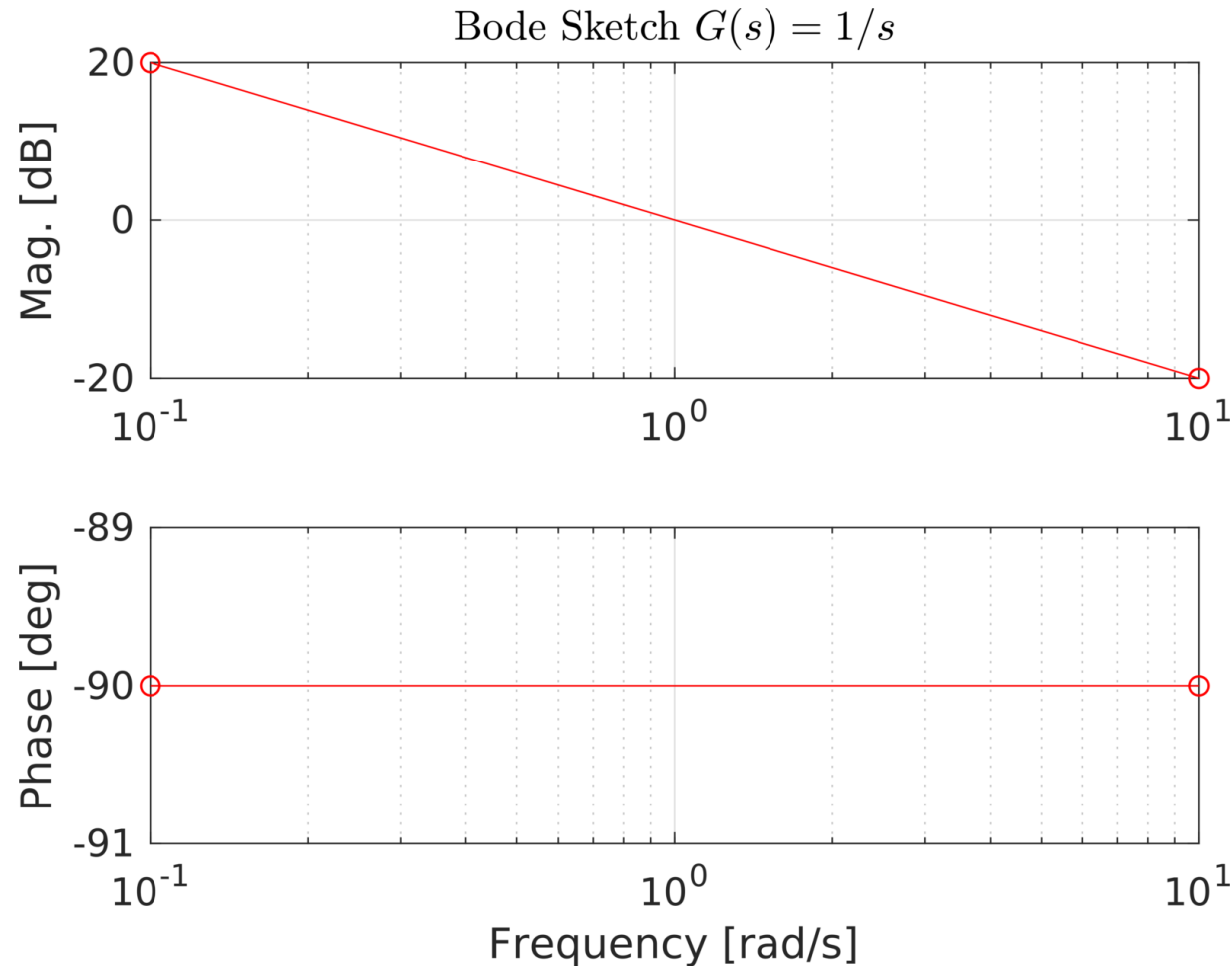
Magnitude

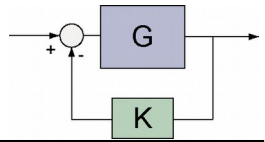
$$M = |G(s = j\omega)| = \frac{1}{\omega}$$

- Slope of -20 dB/decade

Phase

$$\angle G(s = j\omega) = -90$$





Derivatives and Integrals

Order -1 (integral)

$$G(s) = \frac{1}{s}, \frac{10}{s}, \frac{0.1}{s}$$

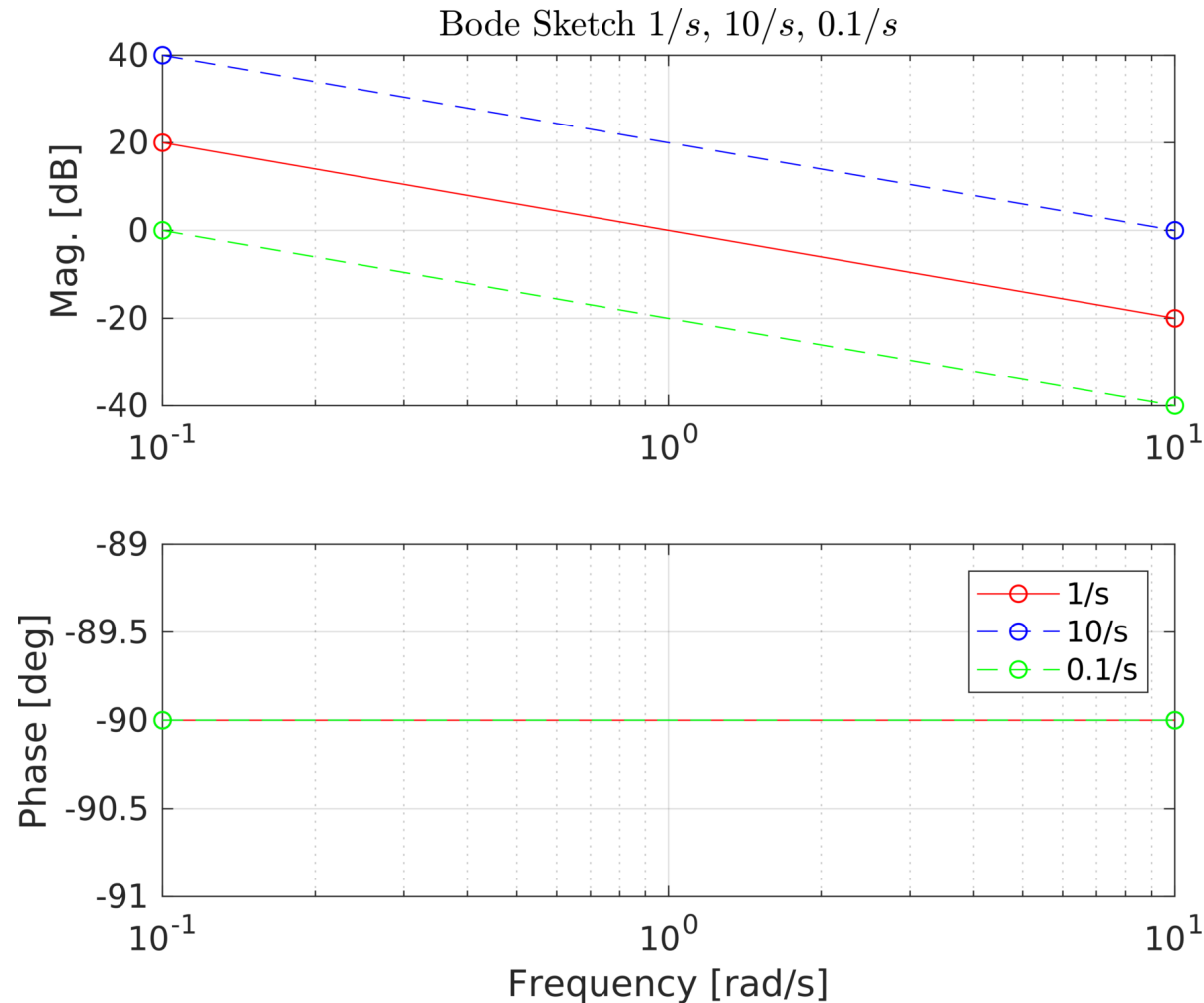
Magnitude

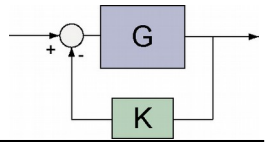
$$M = |G(s = j\omega)| = \frac{1}{\omega}$$

- Slope of -20 dB/decade

Phase

$$\angle G(s = j\omega) = -90$$





Derivatives and Integrals

Order-n (integral)

$$G(s) = \frac{1}{s^n}$$

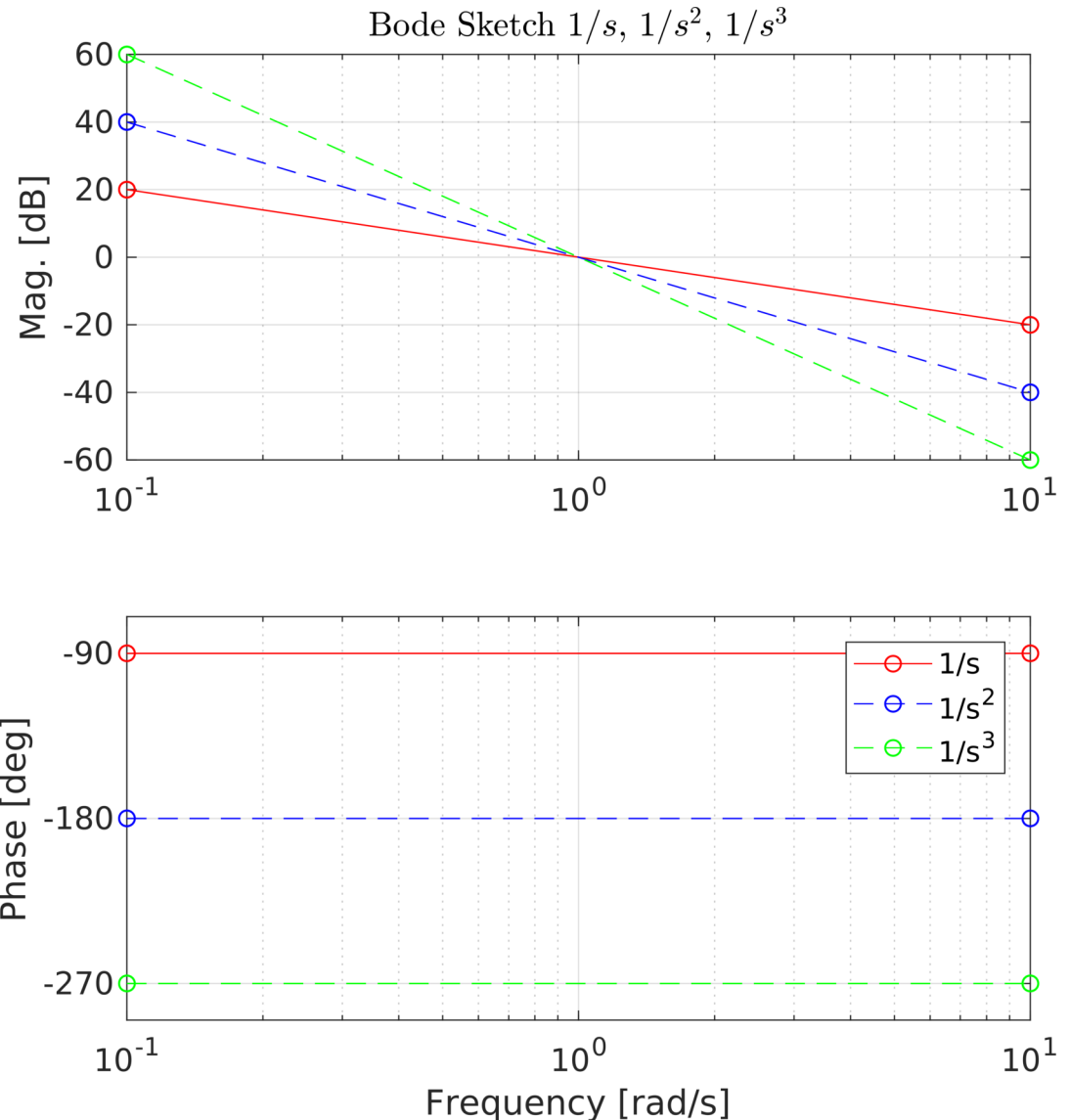
Magnitude

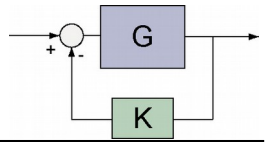
$$M = |G(s = j\omega)| = \frac{1}{\omega^n}$$

- Slope of
-20*n dB/decade

Phase

$$\angle G(s = j\omega) = -90 * n$$





Transfer Function

$$G(s) = (\tau s + 1)^{\pm 1}$$

Numerator: ex. PD

$$G(s) = (\tau s + 1)$$

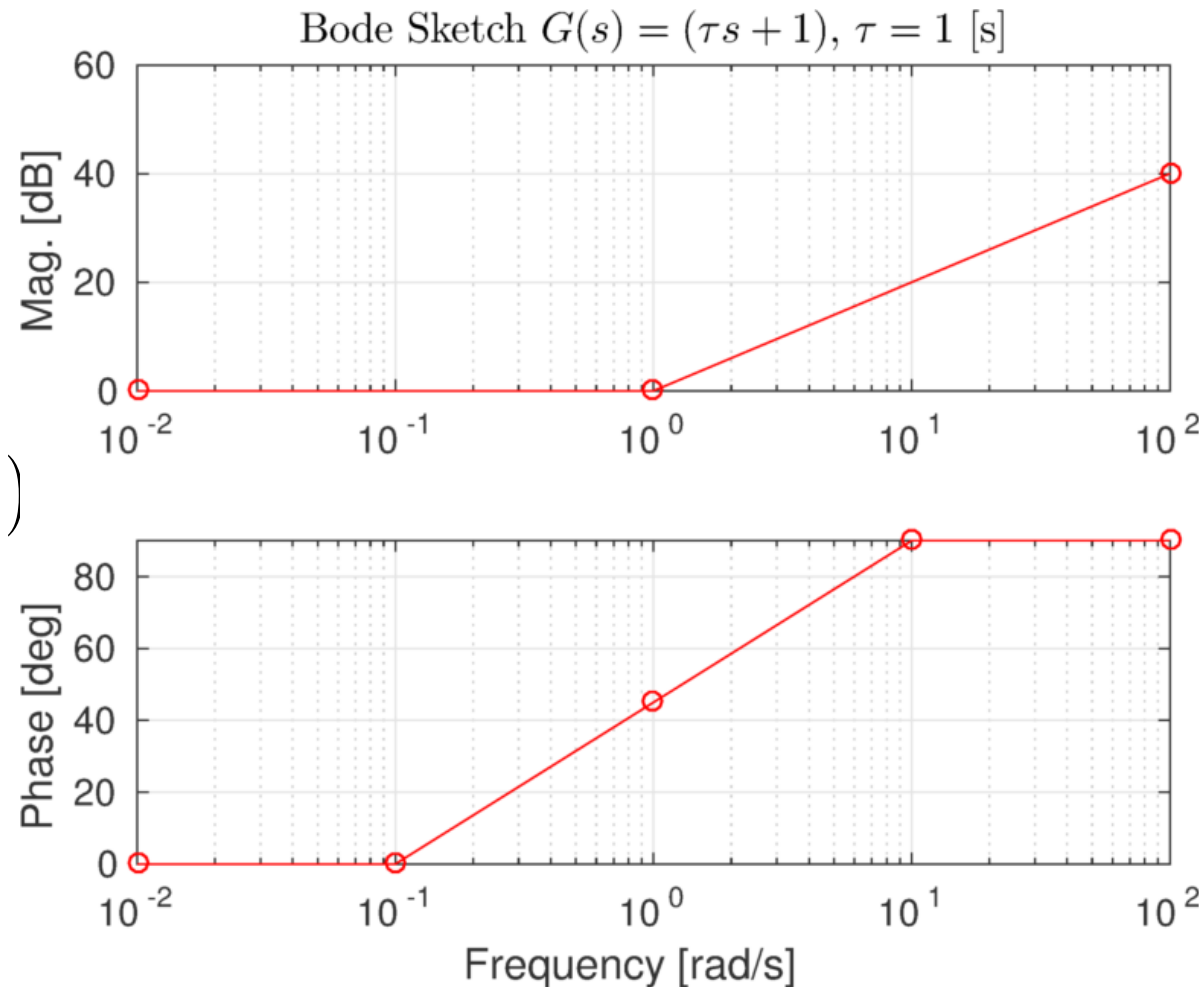
Magnitude

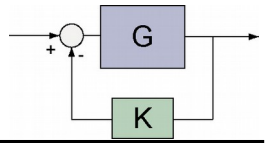
$$M = |G(s = j\omega)| = |\tau j\omega + 1|$$

- Break point @ $\omega_c = 1/\tau$
 $M = \sqrt{2} = 3 \text{ dB}$
- $\omega < \omega_c : M = 1 (0 \text{ dB})$
- $\omega > \omega_c : M = 20 \text{ dB/dec}$

Phase

- $\omega = \omega_c : \varphi = 45$
- $\omega \ll \omega_c : \varphi = 0$
- $\omega \gg \omega_c : \varphi = 90$





Transfer Function

$$G(s) = (\tau s + 1)^{\pm 1}$$

Numerator: ex. PD

$$G(s) = (\tau s + 1)$$

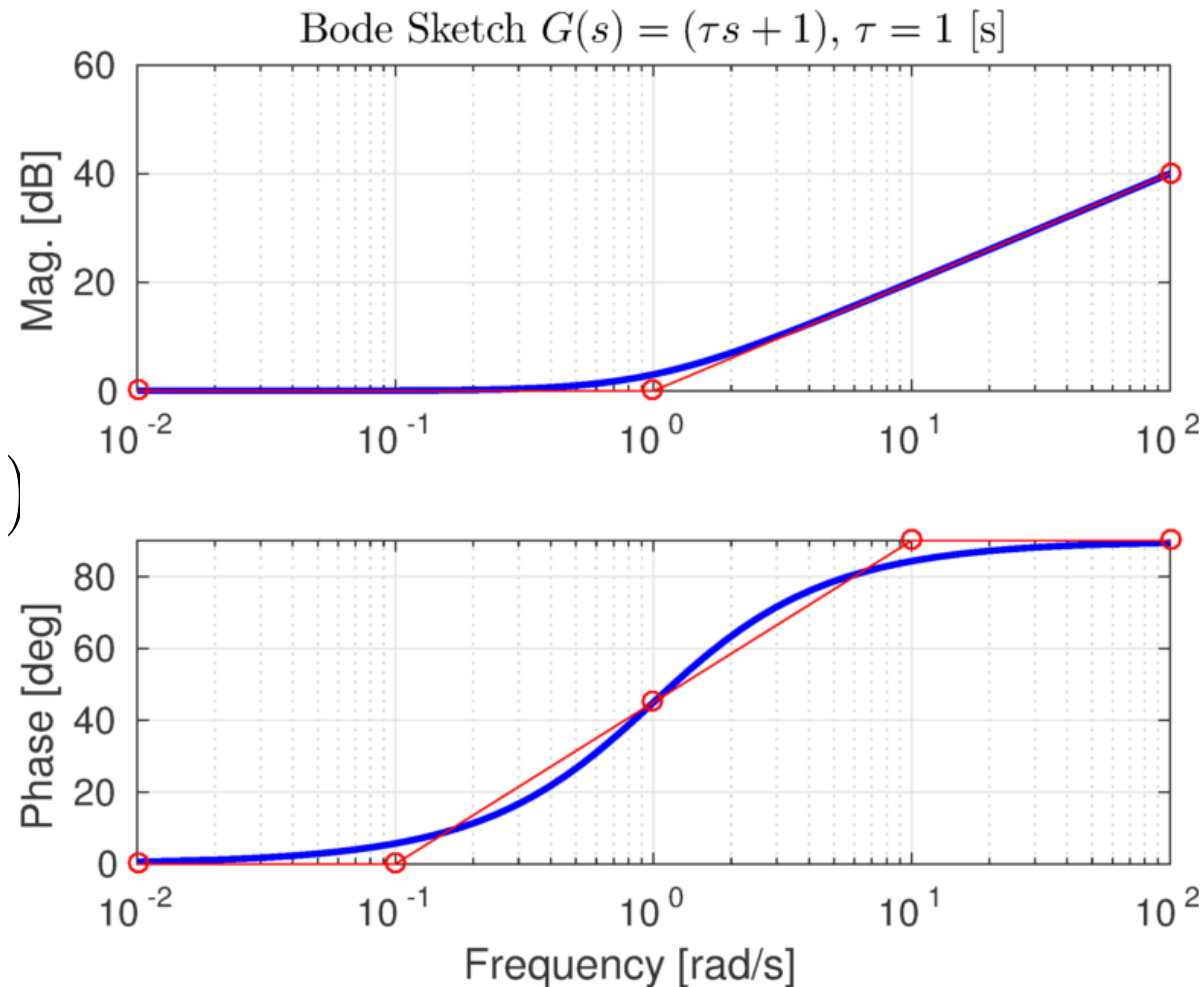
Magnitude

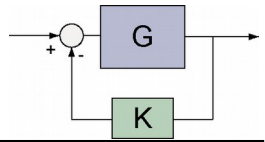
$$M = |G(s = j\omega)| = |\tau j\omega + 1|$$

- Break point @ $\omega_c = 1/\tau$
 $M = \sqrt{2} = 3 \text{ dB}$
- $\omega < \omega_c : M = 1 (0 \text{ dB})$
- $\omega > \omega_c : M = 20 \text{ dB/dec}$

Phase

- $\omega = \omega_c : \varphi = 45$
- $\omega \ll \omega_c : \varphi = 0$
- $\omega \gg \omega_c : \varphi = 90$





Transfer Function

$$G(s) = (\tau s + 1)^{\pm 1}$$

Denominator

$$G(s) = \frac{1}{(\tau s + 1)}$$

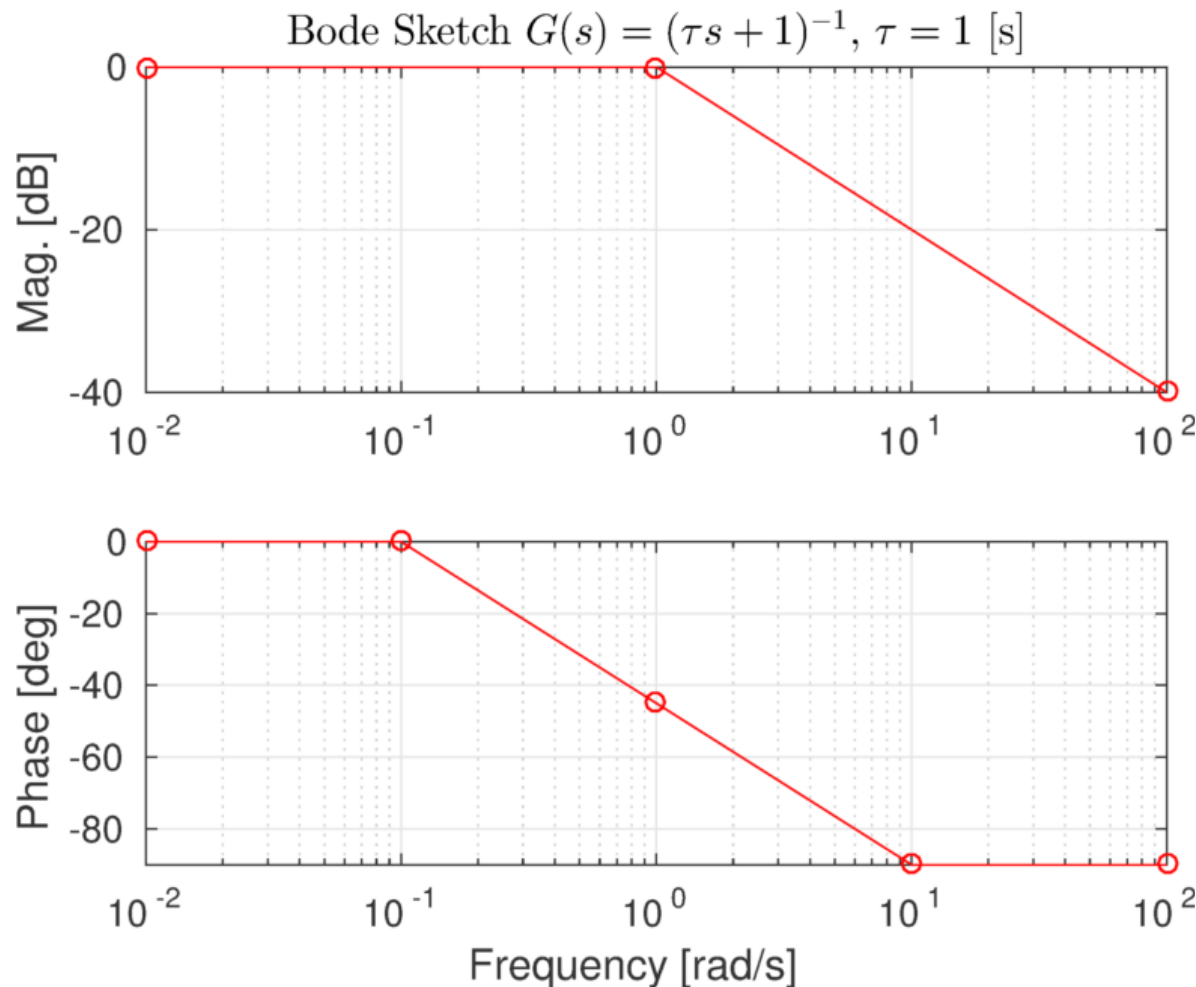
Magnitude

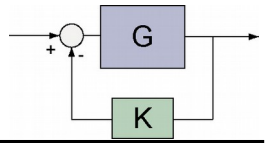
$$M = |G(s = j\omega)| = \frac{1}{(\tau j\omega + 1)}$$

- Break point @ $\omega_c = 1/\tau$
 $M = 1/\sqrt{2} = -3 \text{ dB}$
- $\omega < \omega_c : M = 1 (0 \text{ dB})$
- $\omega > \omega_c : M = -20 \text{ dB/dec}$

Phase

- $\omega = \omega_c : \varphi = -45$
- $\omega \ll \omega_c : \varphi = 0$
- $\omega \gg \omega_c : \varphi = -90$





Transfer Function

$$G(s) = (\tau s + 1)^{\pm 1}$$

Denominator

$$G(s) = \frac{1}{(\tau s + 1)}$$

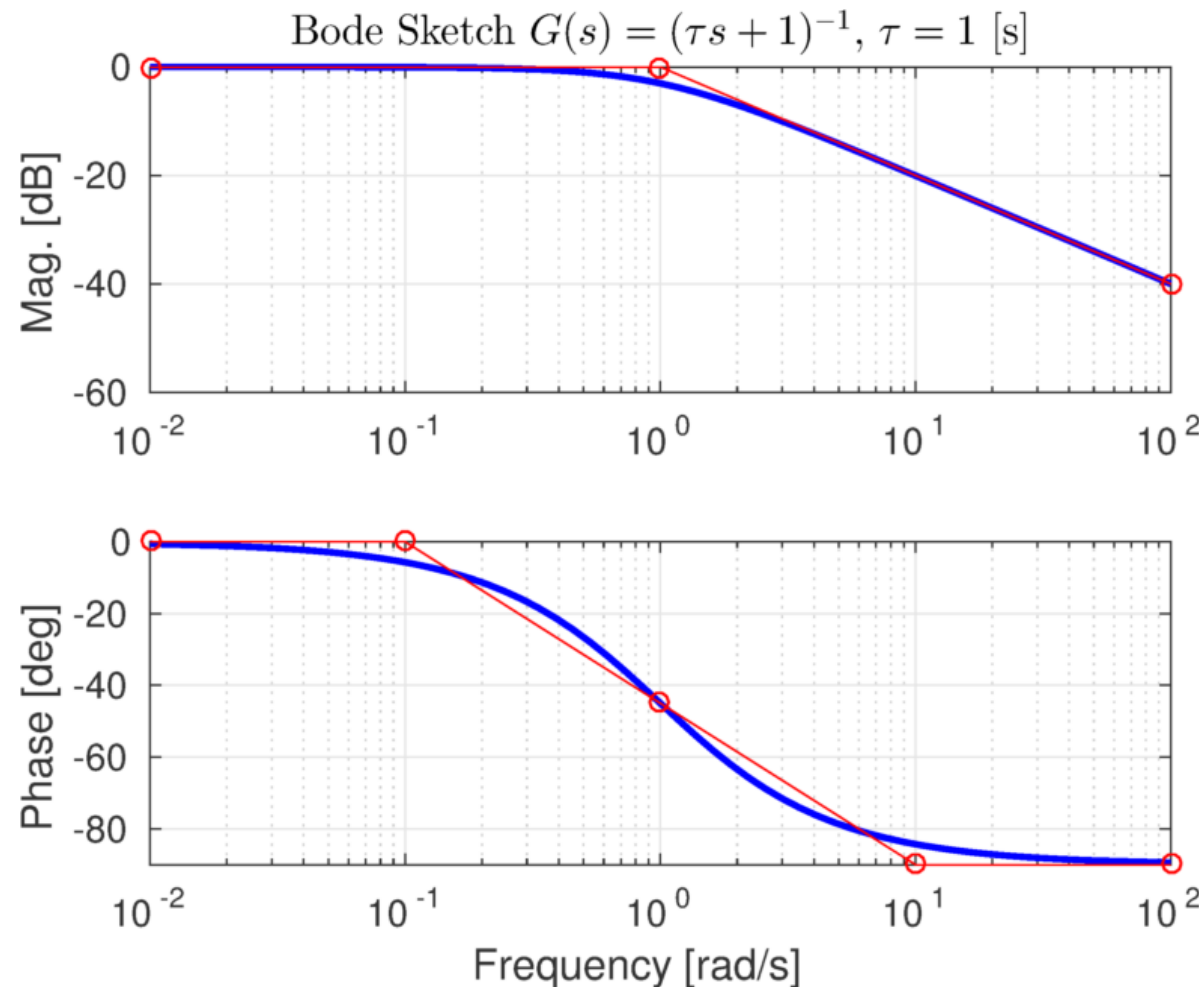
Magnitude

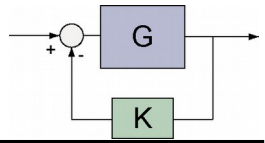
$$M = |G(s = j\omega)| = \frac{1}{(\tau j\omega + 1)}$$

- Break point @ $\omega_c = 1/\tau$
 $M = 1/\sqrt{2} = -3 \text{ dB}$
- $\omega < \omega_c : M = 1 (0 \text{ dB})$
- $\omega > \omega_c : M = -20 \text{ dB/dec}$

Phase

- $\omega = \omega_c : \varphi = -45$
- $\omega \ll \omega_c : \varphi = 0$
- $\omega \gg \omega_c : \varphi = -90$





Transfer Function

$$G(s) = \left(\left(s/\omega_n \right)^2 + 2\zeta s/\omega_n + 1 \right)^{\pm 1}$$

Numerator

$$G(s) = \left(\left(s/\omega_n \right)^2 + 2\zeta s/\omega_n + 1 \right)$$

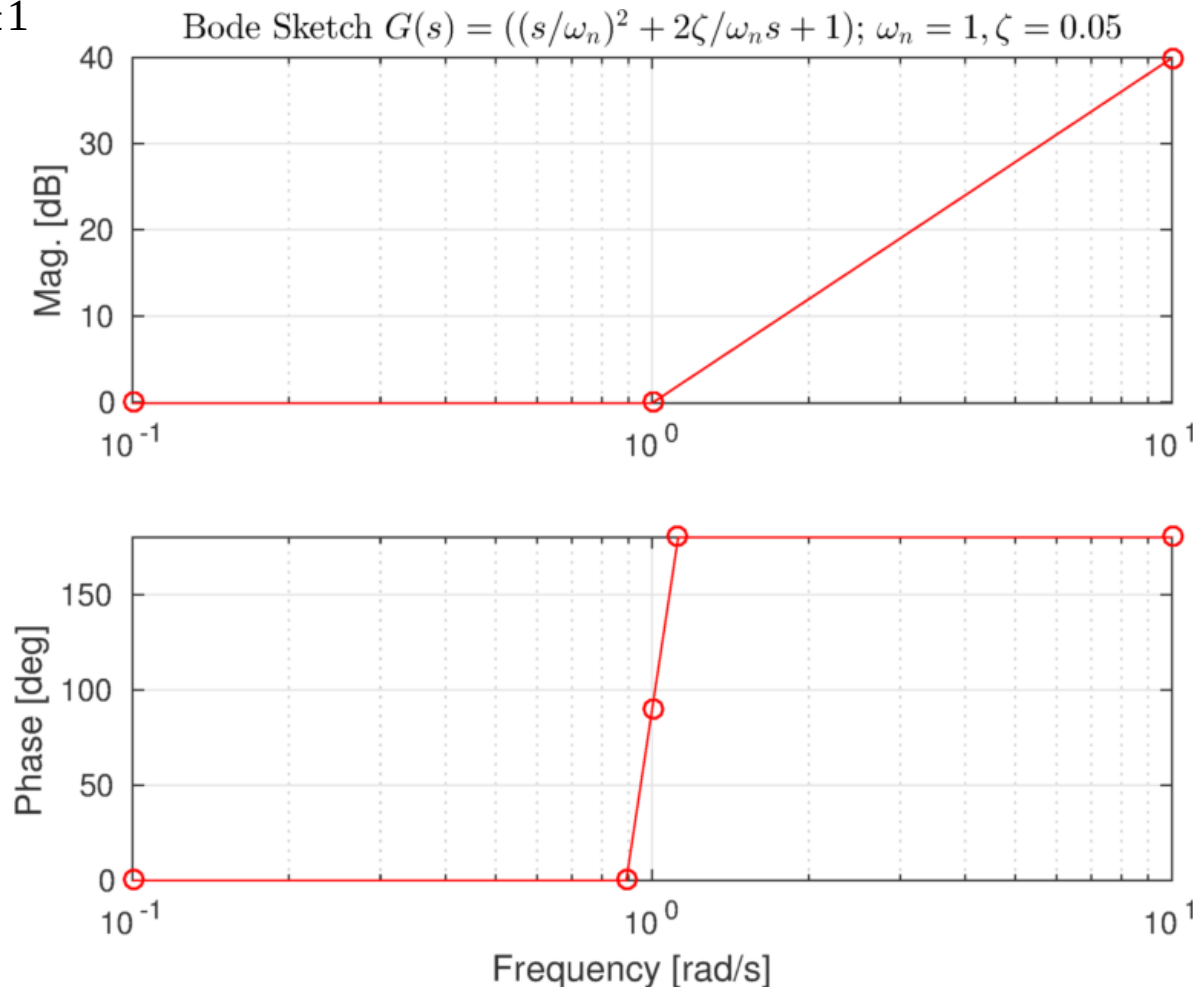
Magnitude

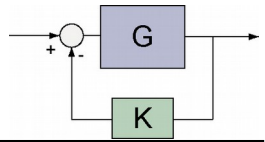
$$M = \left| -\left(\omega/\omega_n \right)^2 + 2\zeta(j\omega)/\omega_n + 1 \right|$$

- Break point @ $\omega_c = \omega_n$
 $M \approx 2\zeta$
- $\omega < \omega_c : M = 1$ (0 dB)
- $\omega > \omega_c : M = +40 \text{ dB/dec}$

Phase

- $\omega = \omega_c : \varphi = 90$
- $\omega \ll \omega_c : \varphi = 0$
- $\omega \gg \omega_c : \varphi = 180$





Transfer Function

$$G(s) = \left(\left(s/\omega_n \right)^2 + 2\zeta s/\omega_n + 1 \right)^{\pm 1}$$

Numerator

$$G(s) = \left(\left(s/\omega_n \right)^2 + 2\zeta s/\omega_n + 1 \right)$$

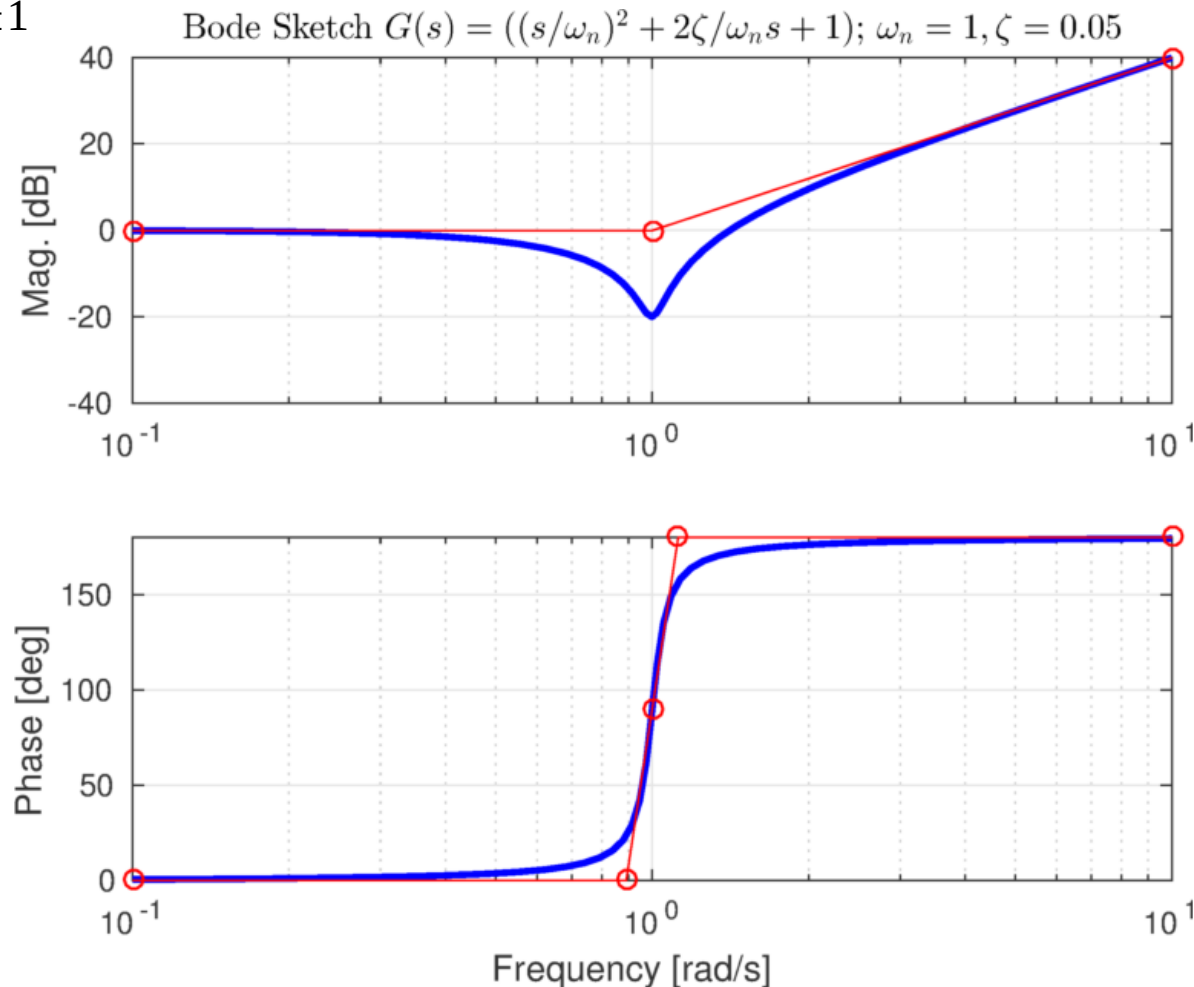
Magnitude

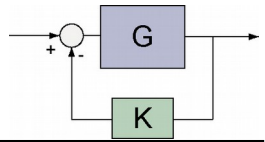
$$M = \left| -\left(\omega/\omega_n \right)^2 + 2\zeta(j\omega)/\omega_n + 1 \right|$$

- Break point @ $\omega_c = \omega_n$
 $M \approx 2\zeta$
- $\omega < \omega_c : M = 1$ (0 dB)
- $\omega > \omega_c : M = +40$ dB/dec

Phase

- $\omega = \omega_c : \varphi = 90$
- $\omega \ll \omega_c : \varphi = 0$
- $\omega \gg \omega_c : \varphi = 180$





Transfer Function

$$G(s) = \left(\left(s/\omega_n \right)^2 + 2\zeta s/\omega_n + 1 \right)^{-1}$$

Numerator

$$G(s) = \frac{1}{\left(s/\omega_n \right)^2 + 2\zeta s/\omega_n + 1}$$

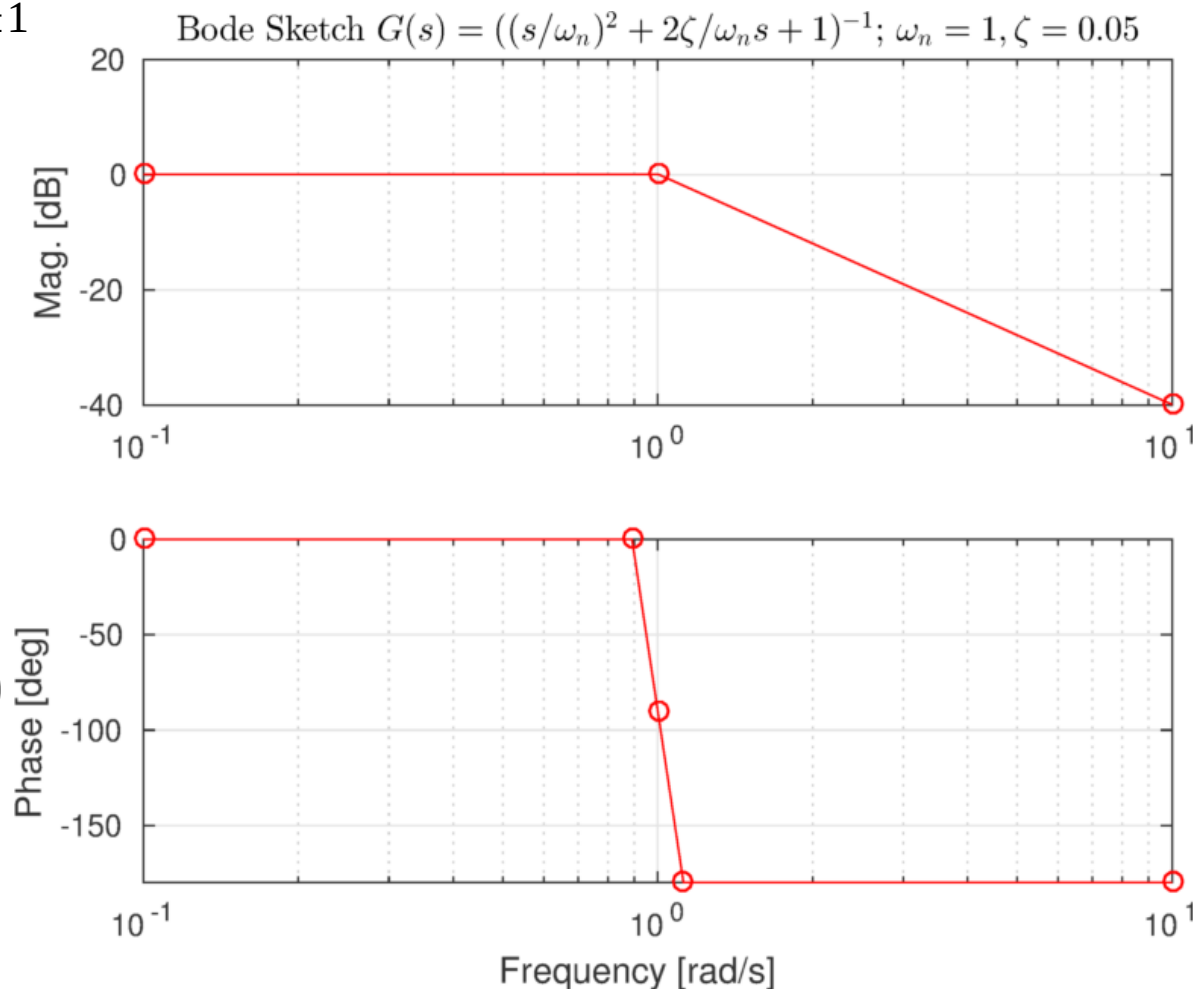
Magnitude

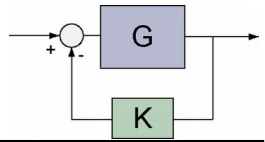
$$M = \frac{1}{\left| -(\omega/\omega_n)^2 + 2\zeta(j\omega)/\omega_n + 1 \right|}$$

- Break point @ $\omega_c = \omega_n$
 $M \approx 1/(2\zeta)$
- $\omega < \omega_c : M = 1$ (0 dB)
- $\omega > \omega_c : M = -40 \text{ dB/dec}$

Phase

- $\omega = \omega_c : \varphi = -90$
- $\omega \ll \omega_c : \varphi = 0$
- $\omega \gg \omega_c : \varphi = -180$





Transfer Function

$$G(s) = \left(\left(s/\omega_n \right)^2 + 2\zeta s/\omega_n + 1 \right)^{-1}$$

Numerator

$$G(s) = \frac{1}{\left(s/\omega_n \right)^2 + 2\zeta s/\omega_n + 1}$$

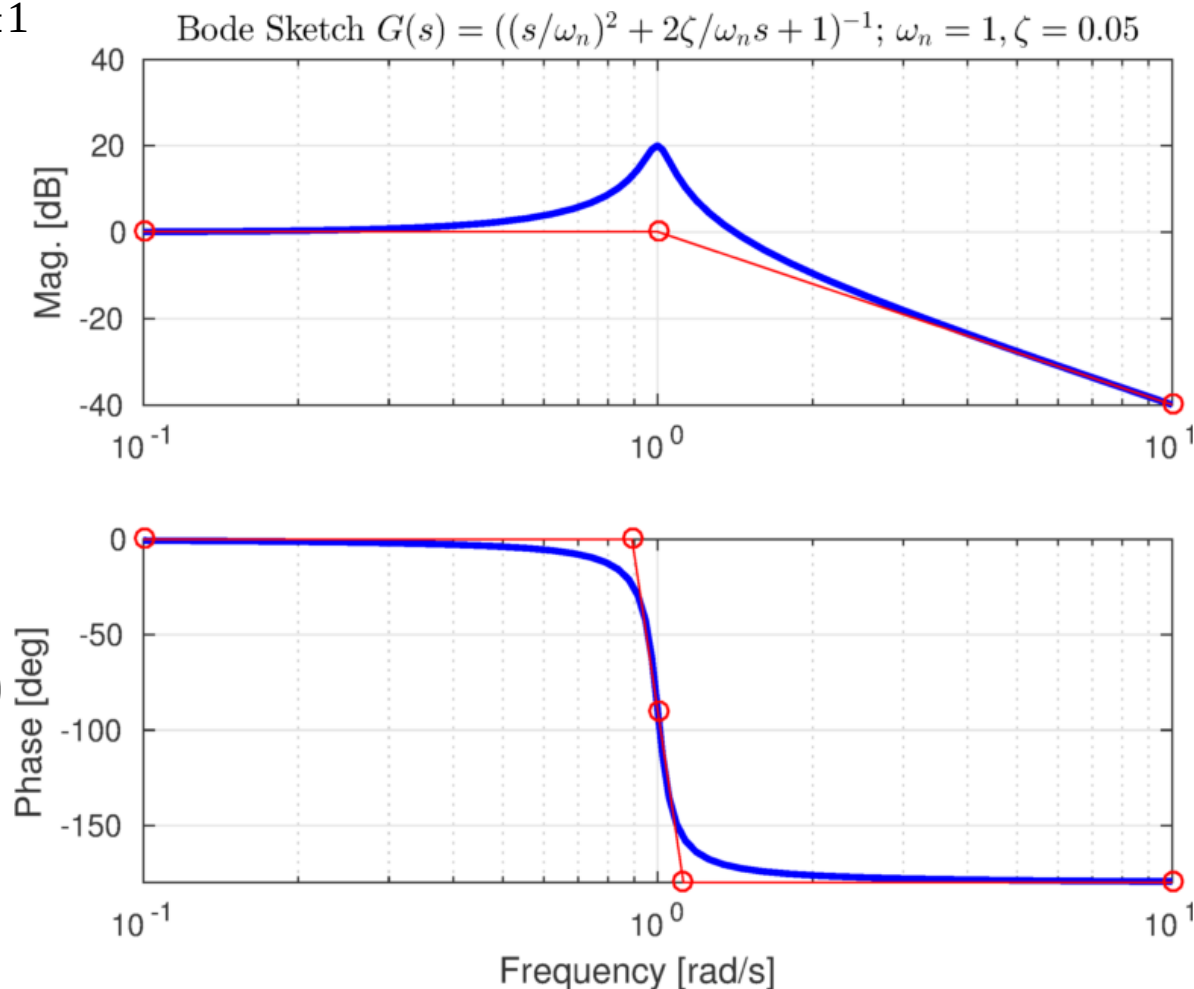
Magnitude

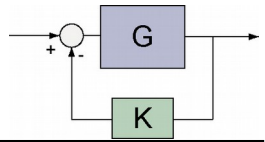
$$M = \frac{1}{\left| -(\omega/\omega_n)^2 + 2\zeta(j\omega)/\omega_n + 1 \right|}$$

- Break point @ $\omega_c = \omega_n$
 $M \approx 1/(2\zeta)$
- $\omega < \omega_c : M = 1 (0 \text{ dB})$
- $\omega > \omega_c : M = -40 \text{ dB/dec}$

Phase

- $\omega = \omega_c : \varphi = -90$
- $\omega \ll \omega_c : \varphi = 0$
- $\omega \gg \omega_c : \varphi = -180$





We can use these parts to estimate the whole
Bode Sketching Process*

1. Put TF into Bode form

$$G(j\omega) = K \frac{[(s\tau_1 + 1)(s\tau_2 + 1)\dots]}{(s)^n [(s\tau_a + 1)\dots][(s^2 + 2\zeta\omega_{n1}() + \omega_{n1}^2)\dots]}$$

2. Separate into building blocks and identify critical frequencies

First order ($\omega_c = 1/\tau$), Second order ($\omega_c = \omega_n$)

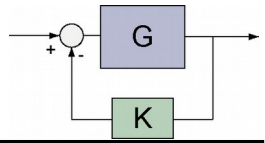
3. Sketch each building block separately

4. Graphically sum the blocks, paying attention to low-freq and high-freq asymptotes. Move from low to high frequency.

* This is a simplified version of textbook process.
Neglect the correction factors at critical frequencies (3 dB, $1/(2*\text{zeta})$, etc.)



Next Time



Examples of Bode Sketching